Practice Questions from Book Chapter 2

2.1 Use algebraic manipulation to prove that $x + yz = (x + y) \cdot (x + z)$. Note that this is the distributive rule, as stated in identity 12*b* in Section 2.5.

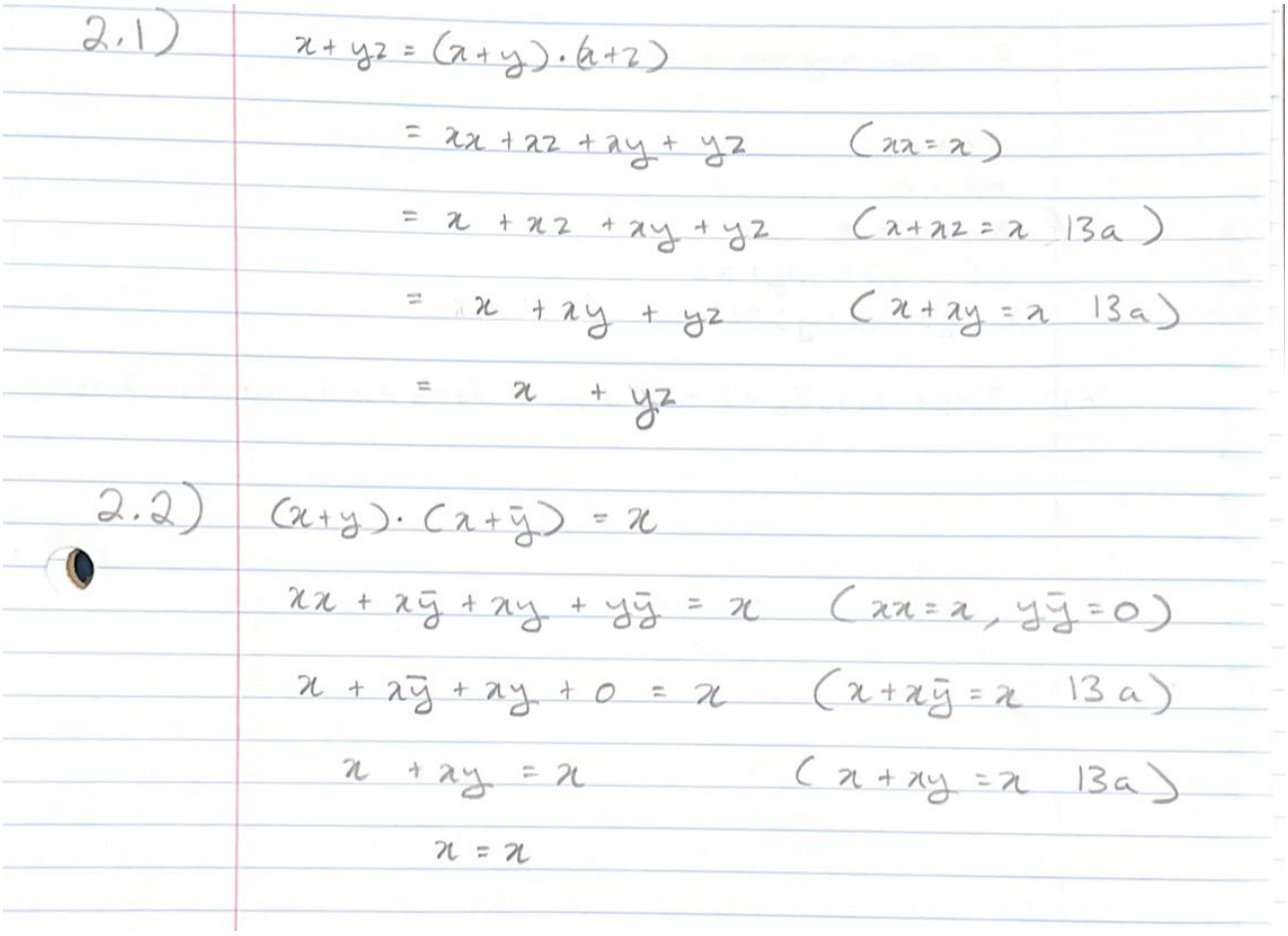
2.2 Use algebraic manipulation to prove that $(x + y) \cdot (x + \bar{y}) = x$.

2.3 Use algebraic manipulation to prove that $xy + yz + \bar{x}z = xy + \bar{x}z$. Note that this is the consensus property 17*a* in Section 2.5.

2.5 Use the Venn diagram to prove DeMorgan's theorem, as given in expression 15*b* in Section 2.5.

2.20 Design the simplest sum-of-products circuit that implements the function $f(x_1, x_2, x_3) = \sum m(3, 4, 6, 7)$.

Solutions:



2.1)

$x + yz = (x+y)\cdot(x+z)$

$= xx + xz + xy + yz \quad (xx = x)$

$= x + xz + xy + yz \quad (x + xz = x \ \ 13a)$

$= x + xy + yz \quad (x + xy = x \ \ 13a)$

$= x + yz$

2.2)

$(x+y)\cdot(x+\bar{y}) = x$

$xx + x\bar{y} + xy + y\bar{y} = x \quad (xx = x,\ y\bar{y} = 0)$

$x + x\bar{y} + xy + 0 = x \quad (x + x\bar{y} = x \ \ 13a)$

$x + xy = x \quad (x + xy = x \ \ 13a)$

$x = x$

2.3) $xy + yz + \bar{x}z = xy + \bar{x}z$

$xy + \bar{x}z + yz = xy + \bar{x}z$

Insert $x + \bar{x} = 1$

Since $x + \bar{x} = 1$, we can write the above expression as

adding $x + \bar{x}$ here makes no difference as $x + \bar{x} = 1$

$xy + \bar{x}z + yz(x + \bar{x}) = xy + \bar{x}z$

$xy + \bar{x}z + xyz + \bar{x}yz = xy + \bar{x}z$

To use 13a, I assume xy as a single variable

$(xy + xyz) + (\bar{x}z + \bar{x}zy) = xy + \bar{x}z$

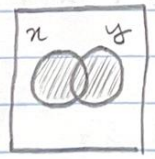
$xy + xyz = xy$ (13a) $\quad \bar{x}z + \bar{x}zy = \bar{x}z$ (13a)

$xy + \bar{x}z = xy + \bar{x}z$

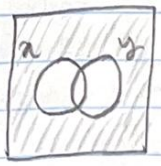
2.5) 15b) $\overline{x+y} = \bar{x} \cdot \bar{y}$

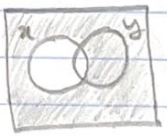
$\overline{x+y}$ $\qquad$ $\bar{x} \cdot \bar{y}$

$x+y$

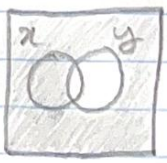


$\overline{x+y}$

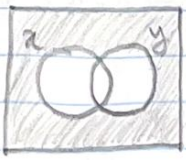




$\bar{x}$



$\bar{y}$



$\bar{x} \cdot \bar{y}$

2.20) $f = m_3 + m_4 + m_6 + m_7$

Row	x_1	x_2	x_3	Minterms
0	0	0	0	
1	0	0	1	
2	0	1	0	
3	0	1	1	$m_3 = \bar{x}_1 x_2 x_3$
4	1	0	0	$m_4 = x_1 \bar{x}_2 \bar{x}_3$
5	1	0	1	
6	1	1	0	$m_6 = x_1 x_2 \bar{x}_3$
7	1	1	1	$m_7 = x_1 x_2 x_3$

$$f = m_3 + m_4 + m_6 + m_7$$
$$= \bar{x}_1 x_2 x_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 x_2 \bar{x}_3 + x_1 x_2 x_3$$
$$= \bar{x}_1 x_2 x_3 + x_1 x_2 x_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 x_2 \bar{x}_3$$
$$= x_2 x_3 (\bar{x}_1 + x_1) + x_1 \bar{x}_3 (\bar{x}_2 + x_2)$$

$$f = x_2 x_3 + x_1 \bar{x}_3$$

